Course Information

- Course: ME 36500 – Measurement and Control Systems I
- Semester: Fall 2025
- Meeting days, times, and locations:
    - Lecture: Tuesday and Thursday
        - CRN-Section – Time – Room – Instructor
        - 11493-077 – 8:30am – ME G061 – C. Chang
        - 29230-059 – 9:30am – ME 2061 – T. Lillian
        - 29231-060 – 11:30am – ME 2061 – T. Lillian
        - 29245-072 – 1:30pm – ME 2061 – X. Deng
        - 29246-073 – 3:30pm – ME 2061 – M. Yuh
        - 12046-097 – 3:30pm – ME G061 – C. Chang
    - Recitation: Thursday
        - CRN-Section – Time – Room – Instructor
        - 11510-100 – 9:30am – ME G061 – C. Chang
        - 11543-104 – 10:30am – ME 2061 – T. Lillian
        - 11511-101 – 12:30pm – ME 2061 – T. Lillian
        - 11514-102 – 2:30pm – ME 2061 – X. Deng
        - 11507-099 – 4:30pm – ME 2061 – M. Yuh
        - 11559-106 – 4:30pm – ME G061 – C. Chang
    - Lab: ME 1042 or ME 1048
- Instructional modality: Face-to-Face
- Credits: 3
- Prerequisites:
    - ME 27400 Basic Mechanics II
    - MA 26200 Linear Algebra and Differential Equations
    - ECE 20001 & ECE 20007 Electrical Engineering Fundamentals I

Instructor Contact Information

- C. Chang
    - Office: ME G092
    - Phone:
    - Email: chang64@purdue.edu
- T. Lillian
    - Office: ME G088
    - Phone: 765-496-7305
    - Email: tlillian@purdue.edu
- X. Deng
    - Office: ME 1069
    - Phone: 765-494-1513
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- M. Yuh
    - Office: NA

    - Phone: NA
    - Email: myuh@purdue.edu
- Office hours will be posted on BrightSpace

Course Description

- Catalog description: The fundamentals of dynamic system modeling are reviewed with special reference to measurement systems. Analytical and experimental techniques of general importance in systems engineering are presented, including sensor utilization in feedback control. Engineering measurement fundamentals, including digital and frequency domain techniques noise and error analysis are covered.
- ME website [https://engineering.purdue.edu/ME/Undergraduate/Files/Courses/me365.pdf]:
    - Title: Systems, Measurements and Control I
    - Course Description: The fundamentals of dynamic systems are reviewed, including modeling of mechanical, electrical, fluid, and thermal systems containing elements such as sensors and actuators used in feedback control systems. Analytical and experimental techniques of general importance in systems engineering are presented. Engineering measurement fundamentals, including digital and frequency domain techniques, noise, and error analysis are covered. Simple proportional controllers are used to convey the benefits of feedback control. Hands-on projects and laboratories are utilized to reinforce fundamental measurement and control system concepts.

Learning Resources, Technology, and Texts

- Learning management system: BrightSpace will be used to share course content, post announcements, and send emails. Students are expected to monitor announcements posted on BrightSpace.
- Gradescope will be used for the submission and grading of assignments and exams.
- Hardware:
    - Computer: Students should have access to a computer capable of running MATLAB® and Simulink® software.
    - ME authorized exam calculator
- Software:
    - MATLAB® and Simulink®: Students will need to use MATLAB® and Simulink® to complete homework and lab assignments.
        - MATLAB and Simulink are registered trademarks of The MathWorks, Inc. See mathworks.com/trademarks for a list of additional trademarks.
        - [https://www.mathworks.com/content/dam/mathworks/mathworks-dot-com/support/books/members/pdfs/mathworks-book-program-brand-guidelines.pdf]
- Textbook: No official textbook is required.
- Supplemental references (available in the library)
    - Print:
        - Palm, William J. System Dynamics. 2nd ed. Boston, Mass: McGraw-Hill, 2010.
        - Figliola, R. S., and Donald E. Beasley. *Theory and Design for Mechanical Measurements*. 3rd ed. New York: Wiley, 2000.

        - Wheeler, Anthony J., A. R. Ganji, and A. R. (Ahmad Reza) Ganji. *Introduction to Engineering Experimentation*. Third edition. Upper Saddle River, N.J: Pearson Higher Education, 2010.
        - Bentley, John P. *Principles of Measurement Systems*. 4th ed. Harlow, England: Pearson Prentice Hall, 2005.
    - Digital:
        - Northrop, Robert B. *Introduction to Instrumentation and Measurements*. 3rd ed. Boca Raton, FL: CRC Press, 2017. [https://purdue.primo.exlibrisgroup.com/permalink/01PURDUE_PUWL/ufs51j/alma99169849919201081].
        - Morris, Alan S., and Reza Langari. *Measurement and Instrumentation: Theory and Application*. Third edition. London, England: Academic Press, 2021. [https://purdue.primo.exlibrisgroup.com/permalink/01PURDUE_PUWL/ufs51j/alma99169717195601081]
        - Blitzstein, Joseph K., and Jessica Hwang. *Introduction to Probability*. Second edition. Boca Raton, FL: CRC Press, Taylor & Francis Group, 2019. [https://purdue.primo.exlibrisgroup.com/permalink/01PURDUE_PUWL/ufs51j/alma99170157757501081]
- Use of artificial intelligence (AI) and/or large language models (LLM) in this course
    - Student use
        - Students are encouraged to use AI and/or LLM tools to study and learn course material.
        - AI and/or LLM tools may be used to provide simple spelling and grammar suggestions for the completion of written assignments.
        - If a student uses AI and/or LLM tools for more than simple spelling and grammar suggestions, the generated material from AI and/or LLM tools is not considered the student's own work. If a student were to present this material as their own, it would be considered academic misconduct. Generated material may be cited following the guidance at: https://www.monash.edu/student-academic-success/build-digital-capabilities/create-online/acknowledging-the-use-of-generative-artificial-intelligence#tabs__3254796-03.
        - Students may not upload, share, or submit course materials (including lecture slides/notes, assignments, exams, lab instructions, and any other proprietary content) to AI tools or LLMs such as ChatGPT, CheggMate, Gemini or similar tools. This restriction applies regardless of whether the content is used for assistance, analysis, or publication. Sharing course materials with AI/LLM tools may violate academic integrity policies, copyright laws, and university data protection guidelines.
        - If you are unsure whether a specific use of AI/LLM is permitted, please consult the instructor before proceeding.
    - Instructor use
        - Instructors may use AI and/or LLM features of Gradescope to assist in grading assignments.

        - Instructors may use AI and/or LLM features of Turnitin to assist in checking the originality of student submitted assignments.
        - Instructors may use AI and/or LLM technologies to generate course content including lecture notes, lecture slides, homework problems, exam questions, etc.
- The Academic Success Center provides services to support Purdue students: https://www.purdue.edu/asc/index.html

Learning Outcomes [https://engineering.purdue.edu/ME/Undergraduate/Files/Courses/me365.pdf]

- Provide a fundamental knowledge of systems, measurements and control.
- Provide an introduction to the concept of feedback control.
- Gain knowledge of the practice and art of modeling, measurements and control through laboratory experiments.
- Provide hands-on experiences with take-home projects and in lab experiments
- Sharpen technical communication skills.

Assignments and Grading Scale

- Final grades will be calculated based on a weighted average of scores on the following assignments:
    - Exams
        - (21%) Exam 1
        - (21%) Exam 2
        - (28%) Final Exam: Time and date to be determined
        - Exams will be closed book and closed notes. A formula sheet will be provided, and students are permitted to use a basic ME approved calculator.
        - All exams are comprehensive.
        - Exams will be graded using Gradescope.
        - For exams 1 and 2, regrade requests must be made through Gradescope within one week of grades being posted to Gradescope. The short timeframe between the final exam and final grade submission deadline prohibits processing of regrade requests for the final exam.
        - As a general rule, **make up exams are not offered**. Alternative exam locations and/or minor changes in exam time may be offered to students with university excused absences. For circumstances in which these accommodations are impractical on exam 1 or 2, the average of the other midterm exam (i.e., exam 2 or 1) and the final exam may be used as the midterm exam score. (At the discretion of the instructor, exceptions to this policy may be granted in rare circumstances that typically result from a major unexpected event.)
    - (7%) Homework
        - About 7 homework regular assignments are expected.
        - Regular homework will be graded more for completion than accuracy.
        - Gradescope will be used for homework submission and grading.
        - As a general rule, **late homework is not accepted**. (At the discretion of the instructor, exceptions may be granted in rare circumstances that result from a

major unexpected event.) Properly submitting homework on time is considered part of the assignment.
    - Keep all records of Gradescope submissions (i.e., keep emails verifying successful submissions to Gradescope).
    - Homework guidelines will be posted to BrightSpace.
- (4%) Simulink Homework
    - 2 Simulink homework assignments are expected.
    - Gradescope will be used for Simulink homework submission and grading.
    - As a general rule, **late Simulink homework is not accepted**. (At the discretion of the instructor, exceptions may be granted in rare circumstances that result from a major unexpected event.) Properly submitting homework on time is considered part of the assignment.
    - Keep all records of Gradescope submissions (i.e., keep emails verifying successful submissions to Gradescope).
- (19%) Laboratory
    - Pre-lab assignments (4%)
    - Lab assignments (15%=2% for each of labs 1-5 and 7 and 3% for lab 6)
    - **To receive a passing grade, students must attend and fully participate in each lab and submit all lab assignments (including pre-lab and lab reports) with a good faith effort.**
    - To attend and participate in lab, students are required to follow all lab safety policies and use common sense to maintain the safety of everyone in lab. If an instructor or lab TA believes a student is being unsafe in the lab, he or she will do one or more of the following: issue a warning, ask the student to leave for the remainder of the lab session, and/or not allow the student to attend lab for the remainder of the semester.
    - All pre and post lab assignments will be submitted using Gradescope.
    - For labs 1-3, a penalty of 30% of the total available points will be imposed for all pre/post lab assignments submitted after the regular due date but before **6pm on October 21**.
    - For labs 4-5, a penalty of 30% of the total available points will be imposed for all pre/post lab assignments submitted after the regular due date but before **6pm on November 18**.
    - A penalty of 90% of the total available points will be imposed for all late pre/post lab assignments submitted after the October and/or November deadlines but before **6pm on December 5**. (At the discretion of the instructor, exceptions may be granted in rare circumstances that result from a major unexpected event.)
    - If a student must miss their scheduled lab due to a university excused event or illness, they must make arrangements with their lab TA (as early as possible) to attend a makeup session. Generally, makeup labs are scheduled with the lab TA to be completed within the same week as the original lab.
    - Students who miss their scheduled lab without an approved excuse will receive a penalty of 90% of the total available points on the associated lab report.

    - (**1% extra credit**) End of semester course survey.
- Details for each assignment will be posted to and submitted using BrightSpace and/or Gradescope.
- Final numeric grades
    - **Grades will be rounded to the nearest whole percentage point** and converted to a letter grade using the following scale
        - A+: 97% to 100%
        - A: 93% to 96%
        - A-: 90% to 92%
        - B+: 87% to 89%
        - B: 83% to 86%
        - B-: 80% to 82%
        - C+: 77% to 79%
        - C: 73% to 76%
        - C-: 70% to 72%
        - D+: 67% to 69%
        - D: 63% to 66%
        - D-: 60% to 62%
        - F: less than 60%
    - Adjustments may be made to account for grading variations due to, for example, exam difficulty and variations between lab or lecture sections.

Attendance Policy

- Students are expected to prepare for, attend, and participate in lecture, recitation, and lab. Students should notify the instructor well before any anticipated absences and as soon as possible for unanticipated absences. Students with university approved absences (including bereavement, military service, jury duty, parenting leave, isolation/quarantine, medical emergencies, and university sponsored activities) will be excused from attendance or given an opportunity to make up the absences. In special cases, and at the discretion of the instructor, accommodations may be offered to students for other absences.
- Students who are required to isolate/quarantine must notify the instructor as soon as possible. As health permits, isolated/quarantined students should continue to submit assignments electronically and stay up to date on course content by reviewing materials posted to BrightSpace.

Course Schedule

- The anticipated course schedule is provided on BrightSpace.

Academic Integrity

- Students are expected to live the Purdue Honor Pledge: “As a Boilermaker pursuing academic excellence, I pledge to be honest and true in all that I do. Accountable together - We are Purdue.”
- Additional information regarding academic integrity may be found at the website for the Office of Student Rights and Responsibilities (https://www.purdue.edu/odos/osrr/academic-integrity/index.html)

- Expectations vary based on assignment
    - Exams: Exams are to be completed individually without any outside assistance or the use of any unapproved sources/technology.
    - Homework: Students are encouraged to work together to complete homework assignments. However, students must fully understand and submit their own work. In addition, particularly useful sources (including discussions with classmates, TAs, instructors) should be cited in the submission.
    - Lab assignments:
        - Calculation/short answer: Students are encouraged to work together to complete calculation/short answer parts of lab assignments. However, students must fully understand and submit their own work. In addition, particularly useful sources (including discussions with classmates, TAs, instructors) should be referenced in the submission.
        - Technical communication: Students must submit their own work for all lab assignments that emphasize technical communication skills (e.g., memo, specification sheet, full lab report, presentation slides). It is unacceptable to submit assignments generated by others and/or artificial intelligence (including large language models and chat bots). However, students are encouraged to seek and incorporate editorial feedback from a variety of sources including, for example, their peers and spelling/grammar tools.
- In this course, credible allegations of academic misconduct will be investigated. In such cases, the instructor will arrange to meet with the students involved. Grading penalties will be determined by the instructor and in consultation with the Associate Head for Undergraduate Studies in Mechanical Engineering when necessary. Grading penalties may include, for example, a zero on the associated assignment(s) and/or a failing final grade for the course. When applicable, grading penalties will be accompanied by a report to the Dean of Students.
- Information about reporting cases of academic dishonesty may be found at: https://cm.maxient.com/reportingform.php?PurdueUniv
- Course materials posted to BrightSpace are protected by copyright law. Most of these materials may be used by students enrolled in the class to assist in learning of course material. However, they may not be shared by any means without the written permission of the copyright holder.

Nondiscrimination Statement

- The BrightSpace page for this course includes a link to Purdue's nondiscrimination policy.

Accessibility

- The Student Services Widget linked on BrightSpace provides a link to the Disability Resource Center (DRC).
- An appropriate Course Accessibility Letter is required for any significant accessibility request. Eligible students should contact the Disability Resource Center (drc@purdue.edu or 765-494-1247) to obtain a Course Accessibility Letter.
- To receive exam accommodations, students must make arrangements with the instructor two weeks prior to any deadlines imposed by the Disability Resource Center and/or Purdue Testing Services. If the instructor and student decide to have the Disability Resource Center or Purdue

Testing Services proctor an exam, the student must have explicit written permission to schedule it for any time that doesn't overlap with the regular exam time.

Mental Health/Wellness Resources

- **Therapy Assistance Online (TAO)** is a self-help webtool available to Purdue students (https://www.purdue.edu/caps/students/resources/self-help/digital/tao.php).
- **Office of the Dean of Students** helps students find many resources across campus (https://www.purdue.edu/odos/ or 765-494-1747).
- **Purdue RecWell** offers personal and group wellness coaching (https://www.purdue.edu/recwell/fitness-wellness/wellness/one-on-one-coaching/wellness-coaching.php).
- **Counseling and Psychological Services (CAPS)** offers a variety of services including professional mental health services (https://www.purdue.edu/caps/index.html or 765-494-6995).
- Additional information and links regarding mental health and wellness are available on the course BrightSpace page.

Basic Needs Security

- The **Office of the Dean of Students** helps students access a variety of resources across campus (https://www.purdue.edu/odos/index.html). Some of these resources are listed below.
- The **Critical Need Fund** helps students overcome unexpected financial emergencies (odos@purdue.edu or 765-494-1747).
- The **ACE Campus Food Pantry** is available to anyone with a Purdue ID (https://www.purdue.edu/vpsl/leadership/About/ACE_Campus_Pantry.html).
- The **Center for Advocacy, Response, and Education** offers services to students who have experienced violence including sexual violence, relationship violence, and stalking (care@purdue.edu or 765-495-2273).
- **Student of Concern Reports** may be submitted to notify the Office of the Dean of Students about students who may benefit from the support and services available on campus (https://cm.maxient.com/reportingform.php?PurdueUniv&layout_id=12).
- Additional information and links regarding basic needs security are available on the course BrightSpace page.

Emergency Preparation

- Sign up to receive PurdueALERT text messages
- The WL Emergency Status page shows the current status of the West Lafayette campus (https://www.purdue.edu/emergency/).
- Periodically review content on the Emergency Preparedness and Planning website (https://www.purdue.edu/ehps/emergency-preparedness/index.php).
- Locate the nearest evacuation route, tornado shelter, and AED
- Be able to distinguish fire alarms and hazard sirens
- Additional information and links regarding emergency preparedness are available on the course BrightSpace page.

Changes to the Syllabus

- This syllabus and the course schedule are subject to change. Updates will be posted to BrightSpace.